Bala Sai Manikanta Sandeep Puppala

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Education

New York University — New York, NY
MS, Computer Science — 09/2026 – Expected 05/2028

Gandhi Institute of Technology and Management (GITAM) — Hyderabad, Telangana, India
B.Tech, Computer Science Engineering | CGPA: 3.79 — 08/2020 – 04/2024

Experience

Storable India — Hyderabad, Telangana, India

Software Engineer I — 04/2025 – 07/2026

- Designed and applied efficient PostgreSQL migrations to add new columns to a production schema, enabling the team to ship on schedule with zero downtime or data loss while reducing deployment risk across 4 production tables.
- Developed an AI-powered chatbot system to generate structured JSON outputs for dynamic section rendering, reducing section creation time from over **2 hours** to under a minute while implementing guardrails for response validation and workflow reliability.
- Built a Micro-Frontend architecture to integrate two web applications into the CMS with a configurable setup, enabling scalable deployment of an improved UI across **600+ facilities**.
- Received the quarterly **Storrior Spotlight Award** four times and the **Emerging Storrior Award** for exceptional performance, rapid growth, and consistent delivery of high-priority tasks.

Associate Software Engineer — 08/2024 – 03/2025

- Led the development of a compliance-related feature by consuming APIs from two products, enabling an on-demand sync every 3 hours and securely storing specific data in the database for seamless website access.
- Managed Redux data storage for site-wide display, preventing potential legal liabilities of **$10,000 per website** for 20% of sites, and implemented secure frontend settings allowing admins to enable or disable the feature.
- Introduced a feature to **70 beta customers**, significantly enhancing the user experience by developing a component to host a web app, triggered after form submission, with a dedicated route and an API for enabling/disabling it via a GraphQL mutation.

Software Development Engineer - Intern — 02/2024 – 08/2024

- Enhanced the CMS UI using the MERN stack, supporting nearly 5000 websites, by upgrading **react-bootstrap** (from 0.32.3 to 2.9.2) and **bootstrap** (from 3.0.0 to 5.0.0), while resolving nearly 50 bugs in 2 weeks and implementing new features along with corresponding backend settings.
- Achieved 1st place in a Hackathon conducted by Storable for developing a Flutter application that provides a seamless experience for the tenants which consumes internal product data via APIs.

Skills

- **Programming**: Python, JavaScript, Java, C, Bash Scripting, OOPs & DSA
- **Web & Backend**: React, Redux, GraphQL, MERN Stack, REST APIs, Micro-Frontend Architecture, Node.js
- **AI & ML**: AI Agents, Prompt Engineering, Scikit-learn, CNNs, Flask, Pandas, NumPy
- **Cloud & Databases**: AWS (S3, Bedrock), Azure Fundamentals, MongoDB, SQL, SQLite, Firebase, Cloud Computing, PostgreSQL
- **Workflow & Tools**: Git, Linux, Agile, Kanban, VS Code, Postman, Android Studio, VIM

Projects

Face-Recognition based Attendance System (Python Flask, Machine Learning) — 06/2023 – 08/2023

- Developed an automated attendance system using a pre-trained **CNN** for face detection, integrated with Python **Flask** for the web interface, and **SQLite** for data management and report generation.
- Evaluated model performance across varying lighting and environmental conditions, identifying accuracy degradation that informed my understanding of deployment challenges in ML systems.

Intelligent System for Public and Private Transportation (Android Java) — 12/2023 – 02/2024

- Developed a multi-user location tracking application using Firebase Realtime Database, enabling users to share their location either 1:1 or 1:n with over 100+ community members.
- Integrated public transportation features, including real-time bus tracking and emergency services like ambulance requests and women's safety alerts based on geo-location.

Research

A Two-warehouse Inventory Model with an Advanced Payment Mechanism for Stock-Dependent Demand, Deteriorating Items with a Fixed Shelf Life and Partial Backlogging — 06/2024

- Developed a mathematical model for managing the inventory of perishable goods with real world complexities.

Achievements

- Ranked amongst Top **3%** candidates in Round I and Top **0.74%** candidates in Round II of IICC (Innovate India Coding Championship) (**1 Lakh** candidates participated) conducted by Chandigarh University, Coding Ninjas and AICTE.
- Secured 96.3469 percentile in JEE MAINS B.TECH.

Leadership & Activities

- **LEAD1x: Exercising Leadership: Foundational Principles** - Harvard University
- Finalist of Aspire Leaders Program Offered by Aspire Institute

Interests: Exploring new technologies by browsing online articles, enhancing my knowledge and staying current with industry trends.